

Recap and Understanding GDP

Session 2 – Introduction to Macroeconomics

Francesco Chiocchio

ESCP Business School

Recap: Session 1

4 minutes. Answer these questions silently, on your own; then we go through the answers.

1. In a model, what is the difference between an **endogenous** variable and an **exogenous** variable?
2. In the circular-flow model, why should **each euro spent** be equal to **each euro earned**?
3. What are the three main **withdrawals** from the circular flow? What are the three main **injections**?
4. If EU households use savings to pay for holidays in the US, is this a withdrawal or an injection for the EU economy?

Recap: Answers I

Questions

1. What is the difference between endogenous and exogenous variables?
2. Why does expenditure equal income in the circular-flow model?

Recap: Answers I

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1. What is the difference between endogenous and exogenous variables?
2. Why does expenditure equal income in the circular-flow model?

Answers

- ▶ **Endogenous variables** are determined inside the model. In the car market model: price and quantity.
- ▶ **Exogenous variables** are taken as given by the model. In the car market model: income or the price of steel.
- ▶ Expenditure equals income because every payment by a buyer becomes revenue to a seller, then income paid to workers, owners of capital, or firm owners.

Recap: Answers II

Questions

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3. What are withdrawals and injections?
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Answers

- ▶ **Withdrawals** are flows of income that leave the domestic circular flow; **injections** are flows of spending that enter it.
- ▶ **Withdrawals/leakages:** saving, taxes, imports.
- ▶ **Injections:** investment, government purchases, exports.
- ▶ Holidays in the US are import expenditure for the EU: money leaves the EU circular flow, so it is a withdrawal for the EU and an export injection for the US.

Session 2: Outline

1. **Why We Care About GDP**

Production, living standards, and human development.

2. **What Is GDP?**

One measure of production, viewed as expenditure or income.

3. **Measuring GDP**

Flows, market values, final goods, and domestic production.

4. **Issues With GDP**

What is omitted, and how to compare countries and years.

5. **GDP From the Expenditure Side**

Consumption, investment, government purchases, and net exports.

6. **How Is GDP Determined?**

Supply, demand, and equilibrium in the long-run model.

Reading: Mankiw, *Macroeconomics*, 12th ed., Chapters 2–4.

1. Why We Care About GDP

Production, living standards, and human development

Why Do We Care About GDP?

GDP is not a perfect measure of well-being, but it is central because it helps us answer:

- ▶ How much is an economy producing?
- ▶ Are living standards improving over time?
- ▶ How do countries compare?
- ▶ Is the economy in a recession or an expansion?
- ▶ How large are the resources available to households, firms, and governments?

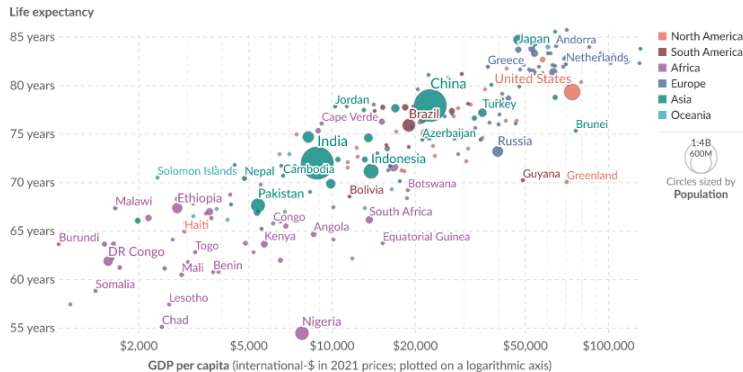
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Main idea: GDP is a measure of production, not a complete measure of welfare.

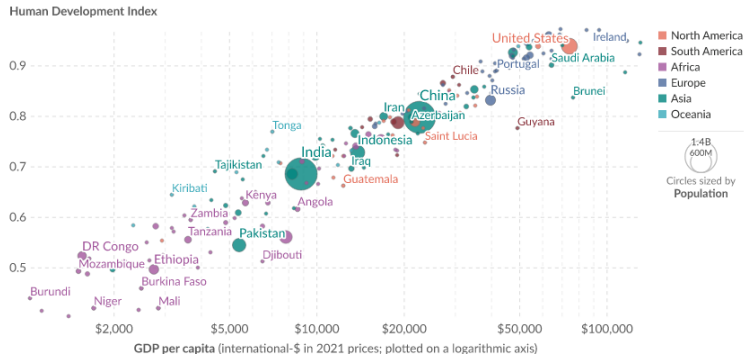
GDP and Life Expectancy, 2023



- ▶ Higher GDP per capita is associated with longer lives.
- ▶ This is a correlation, not proof of causation. Health systems and inequality also matter.

Source: Our World in Data, 2023. GDP per capita is PPP-adjusted; the horizontal axis is logarithmic.

GDP and Human Development, 2023



- ▶ HDI combines health, education, and income.
- ▶ Income is already part of HDI, so the relationship is partly built into the index.

Source: Our World in Data / UNDP, 2023. Standard HDI; PPP-adjusted GDP per capita on a logarithmic axis.

2. What Is GDP?

Production viewed as expenditure and income

Gross Domestic Product

Definition

Gross domestic product (GDP), written Y , is the **market value** of all **final goods and services** produced **within an economy** in a **given period of time**.

1. **Given period of time:** GDP is a flow, usually measured quarterly or yearly.
2. **Market value:** use prices to add different goods and services.
3. **Final goods and services:** avoid double counting intermediate inputs.
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Two equivalent ways to compute GDP

1. **Expenditure approach:** total spending on domestically produced final goods and services.
2. **Income approach:** total income earned by domestically located factors of production.

3. Measuring GDP

From the definition to practical accounting rules

From Definition to Measurement

The definition tells us the practical rules.

1. **Flow:** GDP measures production during a period, not wealth at a point in time.
2. **Market value:** use prices to add very different goods and services.
3. **Final goods:** avoid double counting; use value added when useful.
4. **Produced within an economy:** count production by location.

1. Flow: GDP Is Measured Over Time

Question: Is GDP more like a person's wealth or a person's annual income?

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Answer: GDP is a **flow**: it is measured per unit of time.

Stock	Flow
Wealth	Saving during the year
Capital stock	Investment during the year
Government debt	Government budget deficit
People with college degrees	New college graduates this year

- ▶ GDP is the value of production during a quarter or a year.

1. Flow: Current Production

Used goods are excluded.

- ▶ Resales of cars, furniture, houses, or collectibles are not current production.
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What if unsold bread spoils within the same period?

It adds neither final sales nor closing inventories, so it does not add to GDP. Wages paid to workers are offset by lower profit.

2. Market Value: Adding Apples and Oranges

Nominal GDP

With goods $i = 1, \dots, n$ produced in the period, price P_i , and quantity Q_i :

$$\text{Nominal GDP} = \sum_{i=1}^n P_i Q_i.$$

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Two-good example

Apples (A) and oranges (O):

$$\text{Nominal GDP} = P_A Q_A + P_O Q_O.$$

- ▶ We add apples and oranges by using their market values.
- ▶ Nominal GDP can rise because quantities rise, prices rise, or both.

3. Final Goods and Value Added

Intermediate goods are inputs used to produce final goods.

- ▶ Example: flour used to produce bread.
- ▶ Counting both intermediate goods and final goods would double count.

Production approach: sum value added

- ▶ For each firm:

$$\text{Value added} = \text{Value of output} - \text{Intermediate inputs.}$$

- ▶ Output includes goods added to inventories. Sum value added across producers to obtain GDP.

Example

Farmer $\rightarrow$ Miller $\rightarrow$ Baker:

$$\text{GDP} = \underbrace{(1 - 0)}_{\text{farmer}} + \underbrace{(3 - 1)}_{\text{miller}} + \underbrace{(6 - 3)}_{\text{baker}} = 6,$$

which equals the final price of bread.

3. Final Goods: Imputation

Some production has no explicit transaction price, but it provides services.

1. **Owner-occupied housing:** impute rental value, as if the owner rented the dwelling to themselves.
2. **Certain government services:** value at cost of inputs, especially compensation.
3. **Other imputations:** used where necessary to capture production.

Why impute?

GDP aims to measure current production of final services. When markets are missing, proxies prevent systematic understatement.

4. Produced Within an Economy

GDP counts production by **where it takes place**, not by the nationality of the owner.

- ▶ A car produced in Spain counts in Spanish GDP, even if the firm is foreign-owned.
- ▶ A car produced abroad by a Spanish-owned firm does not count in Spanish GDP.
- ▶ Exports are included because they are domestic production; imports are subtracted because they are foreign production.

GDP and National Accounts

UK GDP by category of expenditure, GNI and NNI: 2015

	£million
Consumption expenditure of households and NPISH (C)	1 216 113
Government final consumption (G)	360 828
Gross capital formation (I)	327 855
Exports of goods and services (X)	510 340
less Imports of goods and services (M)	-548 908
Statistical discrepancy	3 332
Gross domestic product (GDP) (at market prices)	1 869 560
plus Net income from abroad	-37 016
Gross national income (GNI)	1 832 544
less Capital consumption (depreciation)	245 144
Net national income (NNI)	1 587 400

- ▶ **Net income from abroad:** income domestic residents earn abroad minus income foreigners earn domestically.
- ▶ **GNI:** GDP plus net income from abroad.
- ▶ **Depreciation:** the value of capital used up or worn out during production.
- ▶ **Net national income:** $NNI = GNI \text{ minus depreciation.}$

Source: UK Office for National Statistics, *Blue Book 2016*, 2015 data; figure from supplied teaching materials.

4. Issues With GDP

What is omitted, and how to compare countries and years

What GDP Leaves Out

Not all economic activity is captured.

- ▶ **Unpriced services from durable goods:** cars, jewelry, appliances provide services for years, but only their purchase is counted.
- ▶ **Underground economy:** informal or unreported production is hard to observe; statistical agencies estimate part of it.
- ▶ **Non-market household work:** cooking, cleaning, child care, elder care within the family.

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Implication

These omissions make cross-country comparisons harder. If omissions are fairly stable within a country, GDP remains useful for year-to-year comparisons.

- ▶ **Data:** shadow economy by country

Comparing GDP Across Countries Is Hard

GDP is useful for comparing economies, but cross-country comparisons are difficult.

- ▶ Countries use different currencies.
- ▶ Prices differ across countries: the same euro can buy more in one country than in another.
- ▶ The underground economy, household production, and data quality also differ.

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PPP in one line: purchasing power parity adjusts GDP using the price of a common basket of goods, so we compare purchasing power rather than exchange-rate euros or dollars.

- ▶ Because level comparisons are hard, we often compare GDP **within a country over time**.
- ▶ This leads us to focus on **real GDP growth rates**.

GDP Growth Over Time

Why growth? GDP is most useful when we compare production across time.

Real versus nominal growth

- ▶ **Nominal GDP growth:** increase in GDP at current prices, so it captures prices and quantities.
- ▶ **Real GDP growth:** increase in GDP at constant prices, so it captures quantities.

Growth rate of a variable X

$$\text{Growth rate between } t \text{ and } t + 1 = \frac{X_{t+1} - X_t}{X_t} \times 100.$$

- ▶ If real GDP is \$20 trillion in 2022 and \$20.6 trillion in 2023, growth is $\frac{0.6}{20} \times 100 = 3\%$.
- ▶ World Bank GDP growth: China and Italy

5. GDP From the Expenditure Side

Consumption, investment, government purchases, and net exports

GDP From the Expenditure Side

How is the economy's output allocated among different uses?

The national income accounts divide GDP into four broad categories of spending:

- ▶ Consumption
- ▶ Investment
- ▶ Government purchases
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Letting Y stand for GDP:

$$Y = C + I + G + NX.$$

This is the **national income accounts identity**: each dollar of GDP falls into one of these categories.

Consumption (C)

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Consumption is the value of goods and services bought by households.

- ▶ **Durable goods:** cars, home appliances.
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- ▶ **Services:** dry cleaning, air travel, concerts, haircuts.

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Note: New houses are not counted in consumption; they are residential investment.

US Consumption, 2019

	Total, billions of dollars	Per person, dollars
Gross domestic product	21,729	66,199
Consumption	14,795	45,074
Nondurable goods	3,011	9,173
Durable goods	1,548	4,715
Services	10,237	31,186

- ▶ Consumption is the largest component of GDP in the US.
- ▶ Services are the largest part of consumption.

Source: Mankiw, *Macroeconomics*, 11th ed., accompanying lecture slides; 2019 data.

Investment (I)

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No. Investment means spending on new capital and inventories.

- ▶ **Business fixed investment:** plant, equipment, offices, factories.
- ▶ **Residential fixed investment:** new housing units.
- ▶ **Inventory investment:** additions to inventories minus withdrawals, excluding price revaluations.

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Financial investment is an exchange of assets; macroeconomic investment is spending on newly produced capital goods.

What Counts as Investment?

The economy's investment does **not** include purchases that merely reallocate existing assets among different people.

Question: Suppose we observe two events:

- ▶ Smith buys a 100-year-old Victorian house.
- ▶ Jones builds a brand-new contemporary house.

What is total investment here: two houses, one house, or zero?

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Answer: one house.

- ▶ Smith's purchase reallocates existing housing from one owner to another.
- ▶ Jones adds new housing to the economy, so the new house is counted as investment.
- ▶ Investment creates new physical capital that can be used in future production.

US Investment, 2019

	Total, billions of dollars	Per person, dollars
Gross domestic product	21,729	66,199
Investment	3,698	11,267
Nonresidential fixed investment	2,862	8,721
Residential fixed investment	818	2,491
Inventory investment	18	55

- ▶ Investment is much smaller than consumption, but it is central for future production.
- ▶ Inventory investment is often small, but it matters for GDP accounting.

Source: Mankiw, *Macroeconomics*, 11th ed., lecture slides; 2019 data. Nonresidential investment is total investment minus residential and inventory investment.

Government Purchases (G)

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No. Government purchases include spending on goods and services.

- ▶ Examples: trains, subway rail, public schools, national parks, government workers' services.
- ▶ Transfer payments are excluded because they do not directly purchase newly produced goods or services.
- ▶ Examples of transfers: unemployment insurance, pensions, many social benefits.

US Government Purchases, 2019

	Total, billions of dollars	Per person, dollars
Gross domestic product	21,729	66,199
Government purchases	3,814	11,619
Federal	1,450	4,417
Defense	862	2,626
Nondefense	588	1,790
State and local	2,364	7,202

- G measures government demand for goods and services, not the size of the welfare state.

Source: Mankiw, *Macroeconomics*, 11th ed., accompanying lecture slides; 2019 data.

Net Exports (NX)

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$$NX = \text{exports} - \text{imports}.$$

- ▶ **Exports:** goods and services produced domestically and sold abroad.
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- ▶ Imports are subtracted to remove foreign production from domestic spending.

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Interpretation

A trade deficit ($NX < 0$) does not mean “GDP is bad”. It means domestic spending includes more foreign-produced goods than foreign spending includes domestically produced goods.

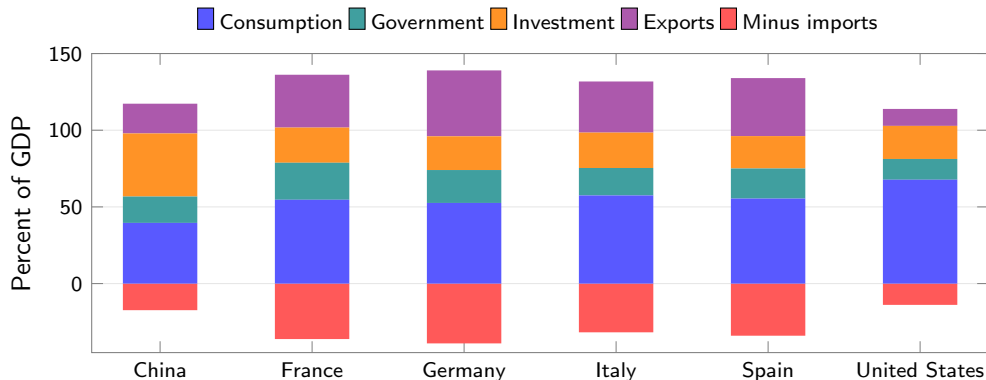
US Net Exports, 2019

	Total, billions of dollars	Per person, dollars
Gross domestic product	21,729	66,199
Net exports	-578	-1,761
Exports	2,498	7,609
Imports	3,076	9,370

- ▶ Imports are not “bad” in the accounting sense; they are subtracted because GDP measures domestic production.

Source: Mankiw, *Macroeconomics*, 11th ed., accompanying lecture slides; 2019 data.

Adding the Components of GDP, 2023



Source: World Bank WDI, 2023. Government is final consumption; public investment is in I (in G in the earlier U.S. tables). Components net of imports sum to roughly 100%, allowing for statistical discrepancies.

6. How Is GDP Determined?

Supply

Demand

Equilibrium

What Determines the Supply of Output?

In the classical model, output is produced with **factors of production** and a **production function**, $Y = F(K, L)$.

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- ▶ L : labor, such as workers' hours.
- ▶ $F(\cdot)$: technology, or how inputs become output.

Bakery example

Capital K	Labor L	Output Y
1 oven	2 bakers	100 loaves
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Constant returns to scale: doubling all inputs doubles output: $F(2K, 2L) = 2F(K, L)$.

Output Supply Is Fixed

For this benchmark model, capital, labor, and technology are given:

$$\begin{aligned}K &= \bar{K}, & L &= \bar{L}, \\Y &= F(\bar{K}, \bar{L}) = \bar{Y}.\end{aligned}$$

- ▶ The available factors and production technology determine output, $\bar{Y}$.
- ▶ The real interest rate adjusts desired spending until it equals this fixed supply.

What Determines the Demand for Output?

Start from the national income accounts identity:

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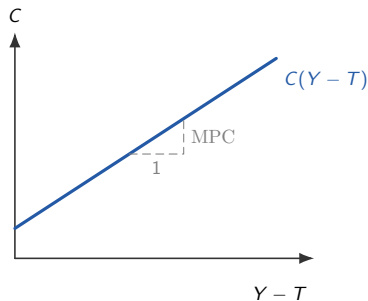
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Demand for output comes from:

- ▶ consumption by households,
- ▶ investment by firms and households buying new capital,
- ▶ government purchases of goods and services.

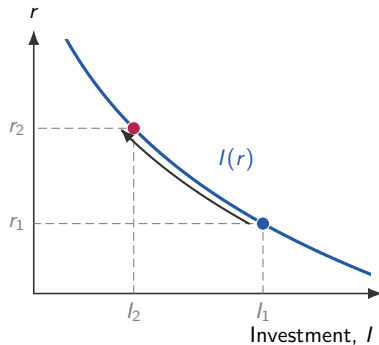
Consumption Depends on Disposable Income

- ▶ Households divide disposable income, $Y - T$, between **consumption** and **saving**.
- ▶ Assume consumption is given by $C = C(Y - T)$.
- ▶ When disposable income rises, C rises.
- ▶ **Marginal propensity to consume (MPC)**: the change in C when $Y - T$ rises by \$1.
- ▶ Assume $0 < \text{MPC} < 1$: households consume part of each extra dollar and save the rest.
- ▶ **MPC will be important for the rest of the course.**



Investment Depends on the Real Interest Rate

- ▶ Firms invest to expand their capital stock and replace worn-out capital.
- ▶ Households invest in new houses.
- ▶ Investment demand depends on the real interest rate r , the cost of funds used to finance investment.
- ▶ A project is profitable if its return exceeds its financing cost.
- ▶ Example: a \$1 million factory returns \$100,000 per year, or 10 percent. Build if $r < 10\%$; forgo if $r > 10\%$.



$$I = I(r), \quad I'(r) < 0.$$

Government Purchases and Taxes Are Policy Variables

- ▶ **Government purchases** are direct demand for goods and services: defense, government employees, schools, roads, libraries, and other public works.
- ▶ **Transfer payments** are different: public assistance and Social Security are not purchases of current output, so they are not included in G .
- ▶ Transfers affect demand indirectly by raising disposable income. We use T as **net taxes**: taxes minus transfers.

In this benchmark model, fiscal policy is treated as exogenous:

$$G = \bar{G}, \quad T = \bar{T}.$$

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$$G = \bar{G}, \quad T = \bar{T}.$$

Equilibrium requires desired spending to equal fixed output:

$$\bar{Y} = C(\bar{Y} - \bar{T}) + I(r) + \bar{G}.$$

Putting Supply and Demand Together

Supply

$$Y = F(\bar{K}, \bar{L}) = \bar{Y}.$$

Fixed in the benchmark model

$$\bar{Y}, \quad \bar{T}, \quad \bar{G}.$$

Demand

$$Y^d = C(\bar{Y} - \bar{T}) + I(r) + \bar{G}.$$

Left to adjust

$$r.$$

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The real interest rate moves until desired spending equals the economy's fixed supply of output.

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Supply of output = Demand for output.

Using the model assumptions:

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- ▶ If desired spending is too high, the real interest rate rises and investment falls.
- ▶ If desired spending is too low, the real interest rate falls and investment rises.
- ▶ Equilibrium pins down the real interest rate.

The Same Equilibrium in Financial Markets

Rearrange goods-market equilibrium:

$$\bar{Y} = C(\bar{Y} - \bar{T}) + I(r) + \bar{G} \quad \Rightarrow \quad \bar{Y} - C(\bar{Y} - \bar{T}) - \bar{G} = I(r).$$

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The left-hand side is **national saving**: output left after consumption and government purchases.

$$\begin{aligned} S &= \bar{Y} - C(\bar{Y} - \bar{T}) - \bar{G} \\ &= \underbrace{\bar{Y} - \bar{T} - C(\bar{Y} - \bar{T})}_{\text{private saving}} + \underbrace{\bar{T} - \bar{G}}_{\text{public saving}}. \end{aligned}$$

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$$\bar{Y} = C(\bar{Y} - \bar{T}) + I(r) + \bar{G} \quad \Rightarrow \quad \bar{Y} - C(\bar{Y} - \bar{T}) - \bar{G} = I(r).$$

The left-hand side is **national saving**: output left after consumption and government purchases.

$$\begin{aligned} S &= \bar{Y} - C(\bar{Y} - \bar{T}) - \bar{G} \\ &= \underbrace{\bar{Y} - \bar{T} - C(\bar{Y} - \bar{T})}_{\text{private saving}} + \underbrace{\bar{T} - \bar{G}}_{\text{public saving}}. \end{aligned}$$

- ▶ **Private saving**: disposable income minus consumption.
- ▶ **Public saving**: net taxes minus government purchases.
- ▶ If public saving is negative, the government is running a deficit.

The Same Equilibrium in Financial Markets

Rearrange goods-market equilibrium:

$$\bar{Y} = C(\bar{Y} - \bar{T}) + I(r) + \bar{G} \quad \Rightarrow \quad \bar{Y} - C(\bar{Y} - \bar{T}) - \bar{G} = I(r).$$

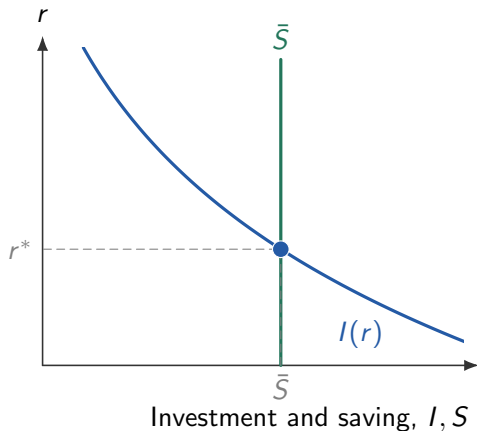
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- ▶ **Private saving**: disposable income minus consumption.
- ▶ **Public saving**: net taxes minus government purchases.
- ▶ If public saving is negative, the government is running a deficit.

So the goods-market condition is also $S = I(r)$.

Equilibrium in the Loanable Funds Market



- ▶ Saving supplies loanable funds.
 - ▶ Why is this supply a vertical line?
- ▶ Investment demands loanable funds.
- ▶ The real interest rate adjusts until

$$\bar{S} = I(r).$$

What to Remember

- ▶ **GDP measures current domestic production.** It is a flow, not wealth, and excludes resales of existing assets.
- ▶ **Production, expenditure, and income describe the same activity.** Value added avoids double counting; $Y = C + I + G + NX$.
- ▶ **Measurement choices matter.** GDP is not a complete welfare measure; compare real quantities and adjust for price differences.
- ▶ **In the long-run benchmark, supply fixes output.** With capital, labor, and technology given, $Y = \bar{Y}$.
- ▶ **The real interest rate adjusts demand.** Goods-market equilibrium is also national saving equal to investment, $S = I(r)$.

Next session: separate prices from quantities and construct measures of inflation.